

Understanding the Effect of Team Performance on Franchise Revenue Under Hard Salary Cap in the NHL

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Abstract

In 2004, the National Hockey League (NHL) experienced trends of declining revenue and severe financial losses among smaller market teams. The league implemented a hard salary cap policy to help encourage competitive balance and resolve this issue. I exploit a difference-in-differences design to estimate how the change in the estimated causal effect of team performance on team revenue differs between large and small market teams after the hard salary cap is implemented. This study will be carried out using team performance statistics from ESPN, Stathead Hockey by Hockey-Reference, and Money puck, city/market demographic data from the US Census Bureau, and franchise financial data from Forbes. Additionally, I plan to investigate how the effect differs within the following three primary streams of sports revenue: broadcasting, arena, and marketing.

Keywords: Hard salary cap, large/small market, team performance, revenue, difference-in-differences

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1 Introduction

Aside from the game, winning, and entertainment, an alternative perspective focuses on professional sports as a business which must ultimately sustain revenue growth and profitability. I will be examining the situation of the hard salary cap in the NHL, highlighting how it impacts the association between team performance and franchise revenue.

The NHL went into a lockout for the entirety of the 2004-05 regular season, meaning there were no games played during this seasonal year due to on-going disagreement in negotiations to update the league's Collective Bargaining Agreement (CBA). The CBA is essentially a textbook of legally binding agreements between the NHL and its representatives for players (NHLPA), including all regulations about revenue sharing, salary cap, and team management. The 2004-05 lockout happened due to larger market NHL teams with highest revenue growth buying up all talented players in the league, causing reduced competition and total revenue in the league. Smaller market teams negotiated for major updates to the new CBA, leading to the introduction of the hard salary cap to encourage competitive balance and higher distribution of talent among teams, thus increasing revenue for small market franchises that were struggling beforehand.

It is understood the rule change had a significant effect on competitive balance and the performance of smaller market teams in the league, but how did this change impact the effect of performance revenue? This focuses on how performance impacts revenue for small and large market teams before and after the hard cap implementation, gaining an understanding for where revenue streams from under each period of treatment. Therefore, my research question is as follows: Does the effect of team performance on revenue change for large-market teams compared to small-market teams after the NHL implemented the hard salary cap in 2005?

In this paper, I use data from ESPN, Hockey-Reference, Moneypuck, US Census Bureau, and Forbes in order to identify the effect of performance on revenue in the NHL at the franchise market size level. I look at the time period between 1995-2015 which allows for ample examination of large market "super teams" pre-2004-05 lockout and cap efficient teams

adjusted to the rule change in the post-period. In order to distinguish large market teams from small market teams, data on local city/market demographics will be used to classify the teams falling into the two groups. Then, the relationship between performance and revenue will be tracked across time for each individual group in a difference-in-differences model, with small market teams being the control group and large market the treatment group. I hope to identify the parallel trends assumption is satisfied with the relationship being similar for each group across all times pre-treatment. Segmenting the teams into market size groups ensures market size is controlled for and that a balanced comparison can be made between the groups.

The rest of the paper is structured as follows. Section 2 provides a review of relevant literature, Section 3 briefly provides additional details on context, policy, and setting, Section 4 provides data I plan to use and why it is well suited, Section 5 outlines my empirical methodology, Section 6 describes my hypothesized results, and finally Section 7 discusses a working plan with next steps to completing the research.

2 Literature Review

Existing literature discusses how team performance can be positively associated with team revenue and profitability from the business operations of professional sports teams. Barajas, Fernandez-Jardon, and Crolley (2005) and Mendes, Goncalves, Henriques, and Tavares (2020) provide evidence of the association between team performance and revenue by highlighting the link in the context of Spanish and Brazilian football. Spanish and Brazilian football are an entirely different sports context from the modern day NHL, with more emphasis on the larger international market and business side. Additionally, international football, like the current MLB and NBA, operate as leagues without the salary cap seen in the NHL, complicating the case of team performance being a cause of revenue. Because of this scenario, the researchers look into the cycle of performance causing higher revenue and higher revenue being reinvested into the team to sustain and grow dominance. The researchers also mention that in this setting, among teams there is a stronger presence of marketing and commercial

based revenue. They also introduce the 3 primary revenue streams in professional sports: arena, broadcasting, and marketing. In regards to empirical methods, they use several tournament and competition types to provide an index to measure team performance and its effect on various segments of revenue.

Nadeau and O'Reilly (2006) develop a profitability model for professional sports teams using the NHL as a case study. They use a data set with over 40 variables, thus introducing numerous predictor variables as viable options to consider in the effect of performance on revenue. Some variables included are ticket pricing, fan income, market size, team history, playoff success, arena seating capacity, etc. The researchers also prove that NHL team winning is particularly associated with higher attendance, merchandise sales, and interest from sponsors. Their empirical method uses the latest season's team ranking and post-season success as indicators of team performance.

Sharma, Treber, and Mulcahy (2016) look into the effect of NHL lockouts. They find evidence that fan attendance is sensitive to the one-time player labor strikes during NHL lockouts, with these events historically decreasing fan demand in the following seasons. Labor disagreements between players and franchise owners have caused NHL lockouts during the 1994-95 and 2012-13 seasons. The NHL lockout in 2004-05 was caused from small market teams calling for a more league-wide balance in competition and revenue, not particularly labor strikes only. However, Sharma, Treber, and Mulcahy make a strong case to investigate the effect of lockouts on lowering fan attendance. This introduces a challenge in identifying the change in the relationship between performance and revenue before and after the 2004-05 lockout.

Fullard (2012) produces research on salary hierarchy in the NHL, focusing on the pay inequality among players on an NHL roster. The study finds that greater pay inequality among players, with the top ranked players on a roster making significantly more, is weakly associated with higher team performance. The study also focusses on goalies at an individual level, finding no evidence of goalie salaries being predicted by goalie performance metrics. Results indicate goalie salaries are more closely related to games played. The lack of evidence

in finding how salary inequality impacts team performance could be because of the NHL's hard salary cap.

Overall, existing literature establishes that performance influences revenue, but the magnitude of this effect is shaped by league institutions and market conditions. There is limited evidence on whether improved team performance translates into higher revenue differently for large- and small-market NHL teams. Compared to global sports such as soccer, the NHL remains more reliant on arena-based revenue streams. Furthermore, there is limited contemporary evidence examining how performance affects revenue in leagues operating under a hard salary cap. Consequently, further research is warranted to evaluate this causal relationship in a modern NHL setting.

This study contributes to the existing literature in three primary ways:

1. It addresses the lack of research on how team performance affects revenue differently across large- and small-market NHL teams by explicitly focusing on market-size heterogeneity.
2. It examines the performance–revenue relationship within a hard salary cap environment, connecting on-ice performance to off-ice financial outcomes while accounting for institutional constraints and market characteristics.
3. It updates an otherwise outdated literature by analyzing the NHL in a contemporary context, during a period of rapid growth in franchise revenues and valuations.

3 Context, Policy, and Setting

Prior to the 2004–05 season, the NHL operated without a hard salary cap, allowing larger market teams with higher revenues to outspend smaller market teams. Because of this disparity, larger market teams were able to offer lucrative contracts and attract more of the talented players, leading to lack of competitive balance in the league. This allowed for large market franchises to build super teams such as the 2001-02 Stanley Cup-winning Detroit

Red Wings, which featured a roster with an astonishing 10 NHL hall of famers. After the 2004-05 lockout, the league implemented the hard salary cap to promote competitive balance, revenue sharing, and more fair revenue distribution across teams. This policy change provides a strong context to investigate the effect of team performance on franchise revenue for small and large market teams with and without the salary cap.

4 Data

Data on NHL team performance statistics such as point percentage, number of wins, and more is sourced from ESPN, MoneyPuck, and Hockey-Reference for the 1994-95 to 2014-15 NHL seasons. This data is well-suited because it is expansive in the statistics offered and is easily accessible for all seasons I intend to study.

Data to evaluate teams' local city and market demographics such as by variables like population, income per capita, age, sex, etc. is publicly available from the US Census Bureau Data website. The data provides variables that can strongly describe NHL franchise markets. Therefore, the data allows for identification and assignment of large and small market teams to their respective groups and provides variables to introduce as controls for market deviation within each group.

Data to provide yearly NHL franchise revenues can be found from accessing Forbes articles, however older data is challenging to find and would require further investigation into older sports business journal releases. Accessing yearly revenue estimates provides values for the outcome variable, but also recording the segmentation of revenue will be even more of a challenge for this study.

5 Empirical Methodology

The most ideal way to investigate the effect of team performance on revenue would be by randomly assigning teams to hard salary cap or no salary cap and investigating the difference in the effect, however this is not possible since all teams operate under the same policies.

My method is to use a difference-in-differences design regression model to investigate the difference in the effect for teams under no salary cap to teams under the newly imposed hard salary cap. I will also want to divide the teams equally into the large and small market groups in order to control for the effect of market size on revenue and investigate the differences in the effect of performance between market sizes. To achieve my difference-in-differences design, I will study each NHL season individually, beginning with 1994-95, and record the effect of performance on revenue using a regression model. All seasons occurring before the 2004-05 lockout will be classified as without the hard salary cap treatment and all seasons occurring afterwards will be classified as with the treatment. Additionally, variables from the US Census Bureau Data will be used to classify and assign teams into the two market sizing groups. Variables such as MSA Population, GDP per capita, sex, and age can be used to create a market index. The franchises ranking in the top 50th percentile of the market index can be assigned to the large market treatment group while the bottom 50th percentile are assigned to the small market control group.

Numerous variables from data from ESPN, MoneyPuck, and Hockey-Reference can be used to create a performance rating index to best evaluate the performance of each team. The index will include point percentage, regulation wins, goal differential, and playoff rounds won.

Control variables will be used to isolate the effect of performance on revenue. Team-level and arena controls include variables such as arena capacity, arena age, arena location within local market/city, ticket prices, expansion team indicator, Canadian team indicator, and new stadium indicator. Market-related controls include variables such as local professional sports competition, age, sex, MSA population, and GDP per capita. Lockout-season indicators should also be included justified by findings from Sharma, Treber, and Mulcahy (2016) previously discussed in Section 2. Team history controls should also be included such as age of the team, number of championships won, number of appearances in Stanley Cup final, and number of players inducted into Hockey Hall of Fame.

Additionally, if data is accessible, the model should be able to produce estimations sepa-

rately for the broadcasting, arena-based, and marketing revenue segments.

My final baseline regression model for the difference-in-differences design can be seen below followed by term descriptions:

$$\begin{aligned}
 \text{Revenue}_{it} = & \beta_1 \text{Performance}_{it} + \beta_2 (\text{Performance}_{it} \times \text{Post}_t) \\
 & + \beta_3 (\text{Performance}_{it} \times \text{LargeMarket}_i) \\
 & + \beta_4 (\text{Performance}_{it} \times \text{Post}_t \times \text{LargeMarket}_i) \\
 & + \gamma_m + \delta_t + X_{it} + \varepsilon_{it}
 \end{aligned} \tag{1}$$

- Revenue_{it} is total (or segmented) franchise revenue for team i in season t .
- Performance_{it} is a continuous performance index capturing on-ice performance.
- Post_t is an indicator equal to 1 for seasons after the 2004–05 lockout.
- LargeMarket_i equals 1 for teams above the 50th percentile of the market size distribution.
- γ_m are market fixed effects
- δ_t are team fixed effects
- X_{it} is a vector of time-varying controls, controlling for league-wide shocks in a given season such as from Lockouts.
- ε_{it} is the error term.

The coefficient of interest, β_4 , measures how the marginal effect of performance on revenue changes for large-market teams relative to small-market teams after the introduction of the hard salary cap.

6 Hypothesized Results

Based on literature and analysis of all factors in my empirical model, I predict that before

the hard salary cap is implemented, performance had a larger effect on revenue for large-market teams due to the performance-revenue feedback loop as discussed in Section 2 from Barajas, Fernandez-Jardon, and Crolley (2005). After the hard salary cap is implemented, I predict that the differences in the effect between large and small markets will shrink due to more league-wide competitive balance and financial constraints. Therefore, overall for both groups, the effect of performance on revenue will weaken after the new cap policy. This is the case because the new setting introduces a significantly more difficult challenge to reach consistently high performance, thus allowing for non-performance factors to be significantly stronger in their effect on revenue. Additionally, regardless of performance, the competitive balance from the cap encourages more equivalent revenues across all teams.

7 Working Plan to Complete Research

In order to carry out the remainder of the research required, I will need to finalize a conglomerate data set for NHL seasons from 1994-95 to 2014-15, including team performance, market demographics, and franchise revenues. I will construct the performance index while classifying teams by market size as planned. The difference-in-differences model will be estimated for total and segmented revenues, with robustness checks using alternative market definitions and performance metrics. I will also verify the parallel trends assumption and interpret results in the context of NHL financial dynamics and policy implications.

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